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Comparison of oxidative treatments of polypropylene films and fibers to promote zinc oxide nanoparticle adhesion

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ABSTRACT

Ozone, heating-in-air, and chromic acid treatments of polypropylene (PP) films and fibers were evaluated for their ability to promote ZnO nanoparticle (NP) adhesion. X-ray photoelectron spectroscopy (XPS) demonstrated that the oxygen content of the surface of PP films increased due to all treatments, with high binding energy O1s components emerging consistent with carbonyl, carboxylic acid, ester and hydroperoxide groups. The latter two were dominant for chromic acid treatment. The XPS measurements were complemented by attenuated total reflectance FTIR spectroscopy studies. Pull-off measurements from PP films, using a ZnO particle attached to the cantilever of an atomic force microscope, indicated that ozone and chromic acid treatments were equally effective at increasing adhesive forces. However, scanning electron microscopy studies of PP fibers treated with either ozone or chromic acid, and subsequent immersion in an ethanolic ZnO NP suspension, showed that ozone treatment was more effective than chromic acid treatment at promoting adhesion. This likely results from the poorer ability of ester and hydroperoxide groups, compared to carboxylic acid ones, to bind to hydroxylated ZnO NPs. It was further shown that excessive ozone treatment leads to brittleness of the fibers, as evidenced by cracking when the fibers are stirred. These results indicate that moderate ozone treatment of PP films and fibers may be a convenient method to attach metal oxide particles to PP surfaces.

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1. Introduction

ZnO is a wide bandgap semiconductor with favorable properties for applications in catalysis [1] and photocatalysis [2], optoelectronic devices [3] and chemical sensing [4]. Exposure of ZnO nanoparticles (NPs) to toxic or hazardous gases produces changes in its fluorescence or electrical resistance [4–6]. This makes nanoparticulate ZnO a promising candidate for use in a hazardous material sensing/catalytic degradation platform.

For these and other applications, it may be beneficial to immobilize the NPs by depositing them on a surface or fabric.

Polypropylene (PP) is one of the most commonly used polymers for fibers due to its mechanical properties and processability. It has been successfully decorated with metal oxide nanoparticles by various methods, including plasma-enhanced chemical vapor deposition [7], sputter deposition [8], hydrothermal synthesis on plasma-treated PP [9], grafting with chemical linkers [10] or atomic layer deposition (ALD) [11–13]. In a previous publication [14], plasma treatment of PP fiber swatches was shown to facilitate the adhesion of ZnO, TiO₂ and Zr(OH)₄ particles *via* the formation of carboxylate groups that covalently bond to the nanoparticles. However, in addition to plasma treatment, other methods are known to oxidize PP [15–31]. These include ozone exposure [18–25] and heating in air [26]. Wet chemical treatments, such as nitric acid, hydrogen peroxide, chromic acid, and a potassium chlorate-sulfuric acid mixture, have also been used [27–31]. A comparison of chemical, electrochemical and UV exposure as a means of “activation” of the PP surface has been reported by Urbaniak-Domagala [27], and chromic acid treatment was found to be extremely effective at oxidizing the surface.

In the present study, the oxidative effects of several surface treatments are studied towards the goal of increasing the adhesion of ZnO NPs to PP films and fibers, with a particular focus on ozonation. While research has been performed related to the effects of ozone on the wetting and adhesive properties of PP [22–25], these have focused on applications such as inks and paints. The present study is the first investigation aimed at optimizing the use of ozone pretreatment for the adhesion of metal oxides to PP.

2. Experimental

2.1. Preparation of PP films and fibers

A 10 wt. % solution of 12 kDa isotactic PP (Sigma Aldrich, stock no. 428116, lot no. MKBX1815V) was prepared by dissolving it in *ca.* 100–120 °C reagent grade o-xylene (Sigma Aldrich); the mixture was heated for 60 min to achieve full dissolution. Silicon wafers, previously cleaned by sonication in acetone, were used as substrates for spin-coating at 2500 rpm for 30 s. The resulting films were then treated using the various oxidation methods, which will be described, and examined by either X-ray photoelectron spectroscopy (XPS) or atomic force microscopy (AFM).

Monofilament fibers were prepared from 250 kDa PP (Sigma Aldrich, stock no. 427888) using a DACA micro-compounder coupled with a co-rotating twin screw extruder operating at 100 rpm, with a 0.75 mm dia. spinneret. A batch of fiber was prepared from 3 g of PP loaded into the hopper that was slowly injected into the micro-compounder barrel operating at 170 °C. As the PP exited the barrel, it flowed out of the spinneret, making a fiber that was collected onto a miniature bobbin. The typical diameter of the fiber was *ca.* 0.2 mm, as measured with an optical comparator.

2.2. Oxidation treatments and ZnO deposition

“Ozone treated” films and fibers were prepared by placing the sample in an Erlenmeyer flask in a fume hood, while ozone was delivered to the flask *via* Teflon

tubing. The ozone was generated by an MKS Semozon AX8407 generator operating at 68% power with 1 L min^{-1} of oxygen gas flow. The ozone output under these conditions is nominally 14 wt. %. “Chromic acid treated” samples were prepared as follows. First, a solution of 2.00 g of potassium chromate (Fisher Scientific) was dissolved in 38 g of distilled water. To this was added 32.13 mL of technical grade sulfuric acid (93–98%, Fisher Scientific), 1 mL at a time, pausing to stir for 15 s after every few ml were added. This resulted in an orange solution. The slow addition was to minimize splashing and evaporation of water from the exothermic dissolution of the acid. The samples to be treated were then submerged in this solution and heated in an oven at 65°C for ca. 50 min before rinsing with distilled water and submerging in distilled water overnight. “Heated-in-air” samples were prepared simply by placing them in a glass petri dish which was placed in an oven at atmospheric pressure and heated at 140°C for 24 h.

Some samples, as will be noted, were immersed in a suspension of ZnO NPs in absolute ethanol and ultrasonicated to show that ZnO NPs remain attached even with mechanical agitation. The ZnO suspension was prepared by ultrasonating 20 mg of ZnO nanospheres (30 nm average particle size, Nanostructures & Amorphous Materials, #5811HT) in 100 mL of absolute ethanol for 1 h. Oxidized fibers or untreated control fibers were then immersed in this suspension and lightly stirred for 2 h before being rinsed with pure ethanol and sonicated for ca. 15 min in ethanol.

2.3. Characterization

XPS was performed with a Thermo VG Scientific Sigma Probe photoelectron spectrometer that used $\text{AlK}\alpha$ X-rays ($h\nu = 1486.7 \text{ eV}$), with a concentric hemispherical energy analyzer operating at a pass energy of 30 eV. The PP coated silicon wafer samples were attached to an electrically grounded sample stage using double-sided copper tape. Photoelectrons were detected at an angle of ca. 37° relative to the sample plane. The C1s and O1s spectral regions were peak-fitted using CasaXPS processing software with a Shirley background. Because PP is insulating, charging occurred due to uncompensated photoelectron ejection from the surface. The C1s spectra were charge corrected by shifting the main aliphatic C1s component to a value of 285.0 eV, which is the generally accepted value for saturated carbons in polymers [32,33]. Oxygen was present in all samples, including the untreated ones, and the O1s spectra were charge-corrected by shifting the lowest binding energy component to 532.6 eV, which is a typical value for oxygen in aged PP samples [33,34].

Attenuated total reflectance (ATR) measurements were performed using a Nicolet iS50 Fourier transform infrared (FTIR) spectrometer. Knitted PP fibers swatches were used, and a background scan was acquired with no sample in the ATR accessory. A resolution of 4 cm^{-1} was used, and 416 scans were averaged for each sample.

The roughness of the spin-coated PP films on Si wafers were measured using a Wyko NT2000 optical profiler. A Bruker Multimode Nanoscope IV was used to perform force-distance AFM adhesion measurements for untreated and oxidized PP. For these studies, an approximately $20 \mu\text{m}$ ZnO particle (Goodfellow Cambridge Ltd., ZN526010/4, $45 \mu\text{m}$ max size) was glued using “J-B Weld Original Cold-Weld” epoxy

to a tipless cantilever (MikroMasch, HQ:NSC35/tipless/AlBS, “C” cantilever with nominal mean 16 N/m force constant). The force constants of the cantilevers were determined by the Sader method [35], which uses the cantilever dimensions, resonant frequency and quality factor. The particle was attached by first carefully dipping the underside of the cantilever into a droplet of the epoxy to wet it, then bringing the cantilever into contact with a particle to pick it up. The epoxy was then allowed to cure overnight.

All force measurements were conducted under nitrogen gas flow in a homemade purging chamber enclosing the AFM to keep the relative humidity below 0.5% to prevent artifacts from moisture. Different locations on each sample were probed, with 50 different measurements taken at each location and averaged together. Results were processed using Scanning Probe Image Processing (SPIP) software to determine the average force of adhesion for all 50 measurements. Since the ZnO particles used do not have a consistent size or shape, the oxidized films were each compared to an untreated film with the same particle/cantilever combination. After all the measurements were collected, the particle on the cantilever was imaged using SEM.

PP fibers were examined using a JEOL JSM 7401 F field-emission scanning electron microscope (FE-SEM). The fibers were coated with ca. 100 Å of gold, using a Denton Vacuum Desk IV sputter coater, prior to introduction into the FE-SEM to reduce sample charging.

3. Results

3.1. X-Ray photoelectron and attenuated total reflectance spectroscopies

C1s and O1s XPS of untreated, ozone-treated (two different exposure times), heated-in-air and chromic acid-treated PP films are shown in Figure 1. The main component at 285.0 eV is assigned to a saturated carbon. A second component at ca. 282.9 eV is due to so-called “inhomogeneous charging” [33]. The third component at 286.2 eV arises from oxidation of the polypropylene and formation of C-O and/or C=O surface species. This component is small for the untreated sample but increases in area for the various oxidation treatments. The oxidized species peaks are relatively small compared to the main (unoxidized) peak because XPS detects 50–100 Å into the sample, greatly exceeding the oxidation depth of the polymer films.

The O1s spectra are more informative because oxygen is mainly present on the surface instead of in the bulk. The dominant component in the spectrum of the untreated film at 532.6 eV is assigned to the oxygen atoms in C=O and/or C–OH. There is also a small component at 534.8 eV. Ozone treatment causes the latter peak, which is ascribed to carboxylic acid groups, to increase in intensity. Prolonged ozone treatment (60 min) causes the higher binding energy peak to increase at the expense of the lower binding energy one, compared to the untreated and 20 min ozone-treated sample. Heating in air and chromic acid treatment also causes the higher binding energy components to dominate and the peaks to increase in binding energy, with observed values of 535.3 and 535.7 eV. These large binding energy values are consistent with the oxygen atoms in ester, peresters, and/or hydroperoxide groups [36].

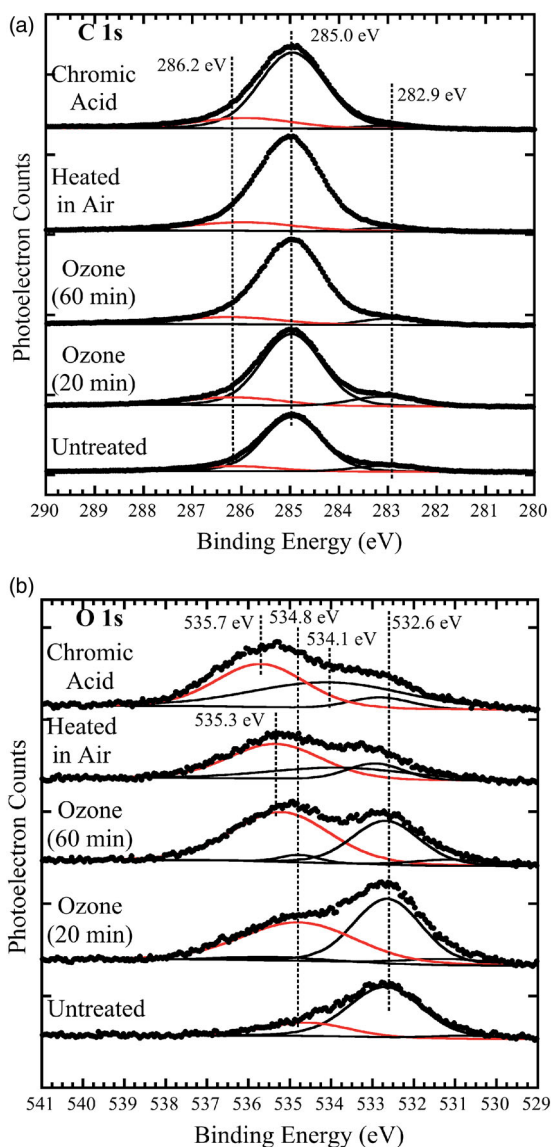


Figure 1. ALK α C1s and O1s XPS spectra of untreated, ozone-treated (for 20 and 40 min), heated-in-air, and chromic acid-treated PP films spin-coated onto Si wafers.

To complement the XPS measurements, attenuated total reflectance (ATR) FTIR spectra were measured of densely knitted PP fiber samples that had been taken through the same treatments as the spin-coated films. Key results are displayed in Figure 2(a,b), which show the carbonyl and hydroxyl group regions, respectively. It is first noted that a small amount of oxidation is observed even for the untreated sample, with a weak peak at $ca. 1738\text{ cm}^{-1}$. This probably results from a small amount of oxidation during the melt extrusion process when the fiber was formed. Heating in air causes this peak to broaden and a small absorbance peak to emerge at 1643 cm^{-1} . Chromic acid treatment results in a peak at 1708 cm^{-1} that accompanies a small 1738 cm^{-1} one. One

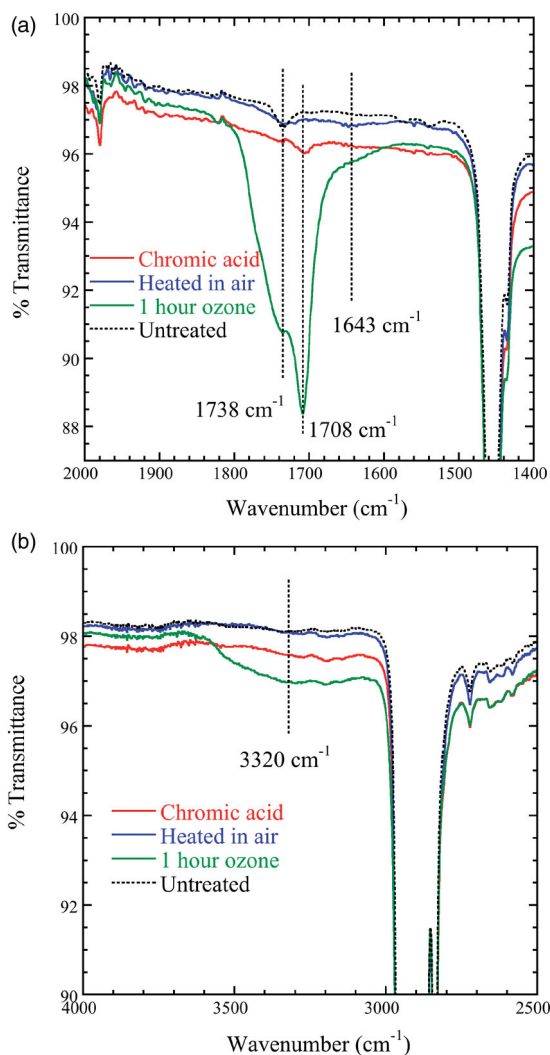


Figure 2. ATR FTIR spectra of (a) the carbonyl and (b) the hydroxyl regions of knitted PP samples after heating-in-air, treatment with chromic acid and 60 min of ozone exposure. A spectrum is also included of untreated PP. There were no noticeable differences in any other regions of the spectra.

hour of ozone treatment induces much more intense and superposed carbonyl peaks. The main component is at 1708 cm^{-1} , with a distinct shoulder at 1738 cm^{-1} . A tail is also present at 1643 cm^{-1} .

It is difficult to make unequivocal assignments of these peaks because multiple species are present. However, comparison of the peak positions to a study by Aslanzadeh et al. [37] that followed the evolution of carbonyl peaks of PP as a function of short wavelength UV degradation time, leads us to assign the 1708 cm^{-1} peak to carboxylic acids and the 1738 cm^{-1} peak to esters and ketones. The small feature at 1643 cm^{-1} has also been interpreted as indicative of hydrogen-bonded carboxylic acid groups [38]. Perester C=O stretches are expected to be observed in the region between $1750\text{--}1785\text{ cm}^{-1}$ [39], but there is no obvious peak in this region for any of the spectra.

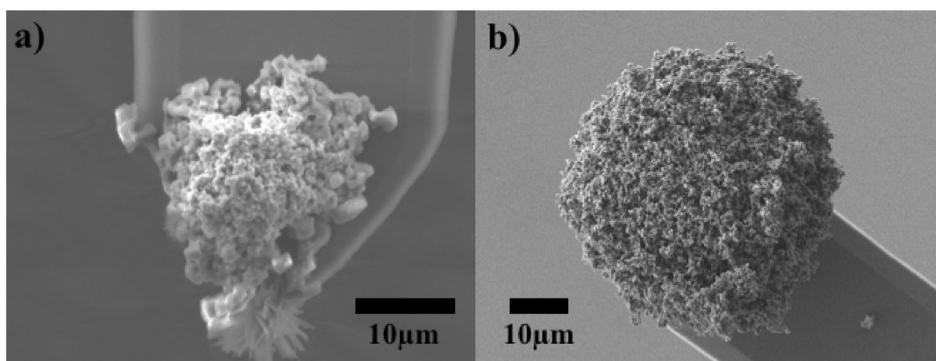


Figure 3. SEM images of ZnO colloidal probes used to collect the AFM data. The probe in (a) was used to obtain the data in Figure 4(a), and that in (b) was used to obtain the data in Figure 4(b). The scale bars in both images represent 10 μm .

Figure 2(b) displays the hydroxyl region, and the very broad peak at *ca.* 3320 cm^{-1} in the ozone-treated sample is likely due to several species, including carboxylic acid groups. Overall, these ATR data indicate that ozone treatment is the most efficient one at forming ketone and carboxylic acid groups.

3.2. Atomic force microscopy measurements

AFM force-distance measurements were performed on untreated, ozone-treated and chromic acid-treated PP film samples. Figure 3 shows SEM images of the ZnO particles attached to cantilevers that were used for these measurements. The particle size is *ca.* 15–30 μm , depending on which particle/cantilever was used. The irregular shape and roughness of the probing particles make it difficult to determine the contact area. However, the same ZnO/cantilever combination was used for the sample of interest and an untreated PP film. In this way, results could be compared by normalizing to the untreated PP sample measurements. Figure 4 shows examples of retraction curves for removal of a ZnO probe from contact with an untreated PP film and a 20 min ozone-treated PP film. For these particular curves, the pull-off forces are *ca.* 750 and 400 nN for the ozone-treated and untreated films, respectively. Corresponding experiments were also performed for a chromic acid-treated sample. As mentioned earlier, multiple measurements were acquired to minimize variations due to differences in polymer film thickness and morphology that cause inconsistent contact areas between the substrate and ZnO particle. The overall results are summarized in Figure 5. Comparison of ozone-treated and untreated PP yields average values of 633 ± 86 and 336 ± 107 nN, respectively. For chromic acid-treated and untreated PP, the values are 87 ± 32 and 54 ± 21 nN, respectively. These results indicate that ozone treatment causes the adhesion force to increase by a factor of 1.9 relative to untreated PP. Chromic acid treatment leads to an increase by a factor of 1.6. These are the same within the errors of the experiment. The smaller values for the chromic acid set of data indicate that the contact area between the probe and sample was smaller for those measurements, even though the overall probe size was actually larger, as shown in Figure 3. This is a

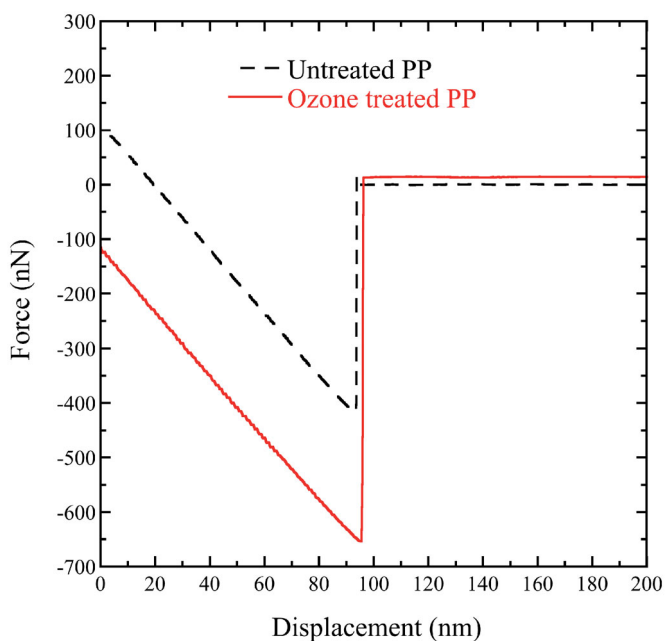


Figure 4. Example retraction force curves for removing a ZnO particle from contact with an untreated PP surface and an ozone-treated PP surface (20 min). For these particular curves, the measured adhesion force was 416 nN and 639 nN for untreated and ozone-treated films, respectively. Both force curves were collected using the same ZnO particle and cantilever.

common feature of these types of measurements in which surface roughness and size of probe asperities dictate the contact area and measured adhesion forces [40].

In order to confirm that differences in surface roughness were not significant factors in the AFM adhesion measurements, the “rms surface roughness” values of the samples were assessed using an optical profiler. The typical thickness of the films was ca. 4 microns. The measured rms roughness values were 203 ± 9 , 250 ± 25 , 180 ± 6 , and 256 ± 38 nm for the untreated, heated-in-air, chromic acid-treated, and ozone (1 hr) treated samples, respectively. While chromic acid etching has been shown to increase the surface roughness of some PP-containing polymers (e.g. ethylene-PP block copolymers or rubber-modified PP), the increase in surface roughness of PP homopolymers has been shown to be minimal [29].

3.3. ZnO deposition

In order to assess the ability of the oxidation treatments to affect the adhesion of ZnO NPs to PP, an ozone-treated film (on a Si wafer) was immersed in the ethanolic ZnO NP suspension, along with an untreated film. Both samples were gently washed with pure ethanol and dried in air. They were not ultrasonicated, because this was found to cause the film to delaminate from the Si wafer. The samples were then introduced into the XPS instrument and analyzed. The results are shown in Figure 6. The presence of Zn2p signal in the ozone-treated sample spectrum demonstrates that the treatment

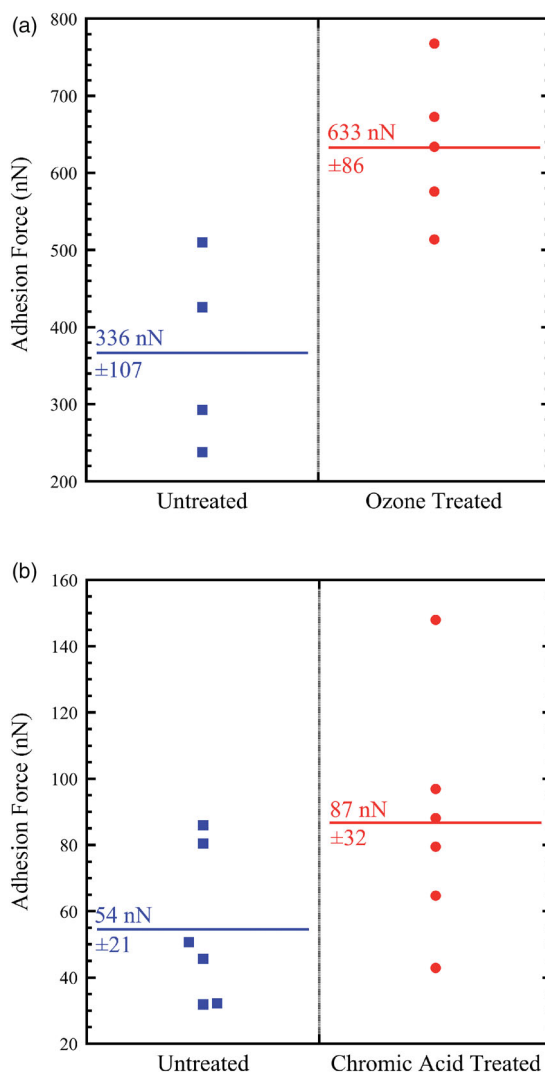


Figure 5. Colloidal probe AFM results for the comparison of (a) untreated and ozone exposed PP films and (b) untreated and chromic acid exposed films. Each point is the average of 50 force curves collected at nominally the same location on the film, and the horizontal line is the average of the data points taken at different locations.

facilitates greater NP attachment. The Zn/C atomic ratio for the ozone-treated and untreated films are 0.118 and 0.011, respectively.

Similar experiments were performed on PP fibers by immersing them in a ZnO NP suspension, rinsing them in pure ethanol and then ultrasonicing them for *ca.* 15 min. FE-SEM was performed instead of XPS to assess ZnO adsorption. The latter is not feasible, with our instrumentation, because of the small size of the fibers. The SEM results are presented in Figure 7. The presence of a ZnO coating is evident in the image of the 60 min ozone-treated fiber. Interestingly and somewhat surprisingly, chromic acid treatment does not lead to ZnO deposition. Possible reasons for this will be discussed shortly. In the case of the ZnO-coated ozone-treated sample, a crack in the coating is

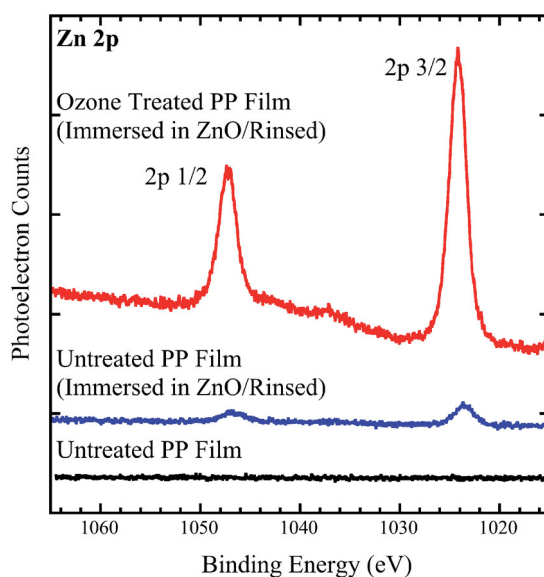


Figure 6. Alkali Zn2p XPS spectra of an untreated PP film, an untreated PP film immersed in colloidal ZnO and then rinsed, and a 20 min ozone-treated PP film immersed in colloidal ZnO and then rinsed.

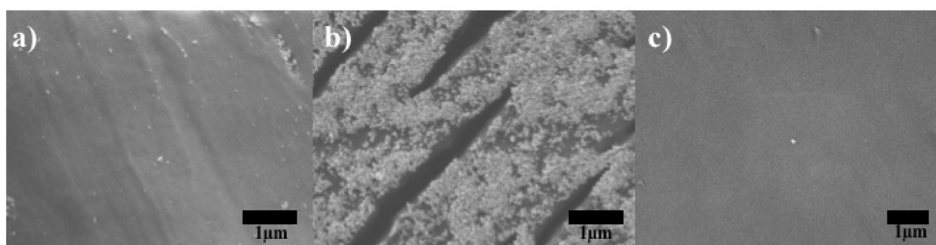


Figure 7. FE-SEM images of PP fibers that were stirred in a ZnO NP suspension, rinsed and then ultrasonicated. Prior to this, the fibers were: (a) untreated, (b) ozone-treated for 60 min, and (c) chromic acid-treated.

obvious. To further investigate the origin of this crack, an experiment was performed in which PP fibers that had been ozone treated for 20 and 40 min were stirred in pure ethanol, in a similar way as to when ZnO was present. FE-SEM images are displayed in [Figure 8](#) and demonstrate that negligible cracking is observed for 20 min of ozone. However, after 40 min, cracks are obvious, as exemplified in the figure. These results indicate that the crack observed in [Figure 6](#) is due to the fiber being weakened by excessive ozone. It is, therefore, believed that the crack in the fiber leads to the crack in the ZnO coating that is on top of it.

4. Discussion

The types of species produced by oxidation of PP are shown in [Figure 9](#). These include alcohols, hydroperoxides, carbonyls, carboxylic acids, esters and peresters [29,41,42]. In general, oxygen atoms in more electronegative environments, such as when bound to

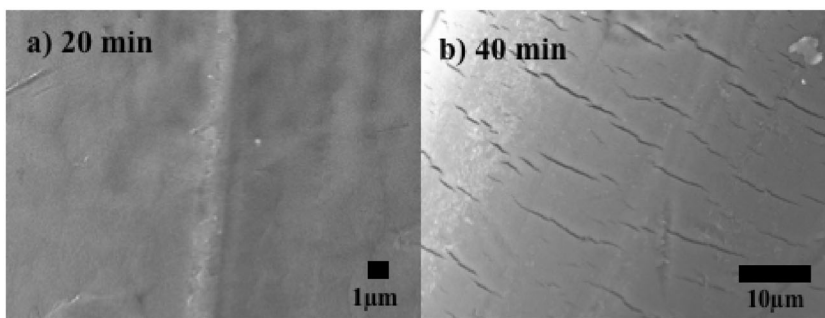


Figure 8. FE-SEM images of PP fibers that were ozone-treated for (a) 20 min and (b) 40 min prior to stirring them in pure ethanol. The cracks obvious in figure (b) suggest that longer ozone treatment leads to compromised fiber strength through polymer backbone cleavage.

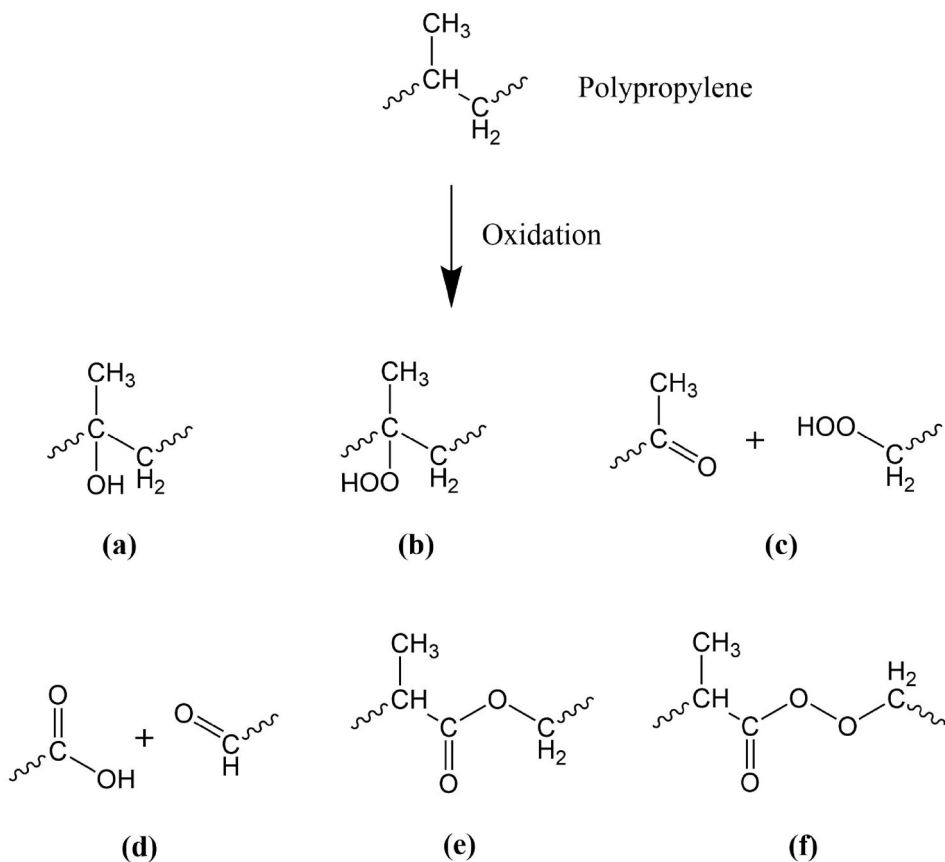


Figure 9. Examples of the types of products that result from oxidation of polypropylene: (a) alcohols, (b) hydroperoxides, (c) carbonyls and hydroperoxides, (d) alcohols and carbonyls, (e) esters, and (f) peresters.

other oxygen atoms, give rise to higher O1s binding energies. The XPS spectra in Figure 1 suggest that ozone treatment is less aggressive with respect to oxidation than either heating-in-air for 24 h or chromic acid treatment.

Ozone treatment mainly leads to the formation of carbonyl and carboxylic acid groups, and the latter can react with the ZnO NPs, which are essentially fully hydroxylated [6], *via* a condensation reaction. In contrast, esters, hydroperoxides, and peresters formed by heating PP films and fibers in air and by chromic acid oxidation are not suitable for covalent bond formation with the ZnO NPs. This is consistent with the PP fiber immersion studies in colloidal ZnO that show that ozone treatment is effective at promoting ZnO adhesion. The ATR and XPS data do not fully agree, likely due to the differences in detection depth. XPS is significantly more surface sensitive, and it is likely that species such as hydroperoxides may be on the very top surface of the chromic acid-treated sample but not below the surface.

A negative aspect of ozone treatment, however, is that it compromises the mechanical properties of the fibers, especially for extended ozone exposure, as evidenced by the cracks revealed by SEM in Figures 7 and 8. These occur because of the scission of the polymer backbone, as illustrated, for example, by the oxidation reactions shown in Figure 9(c,d).

At first glance, the AFM pull-off data seem inconsistent with the ZnO NP immersion studies. The AFM measurements indicate essentially equal pull-off forces (when normalized to untreated PP) for the ozone-treated and chromic acid-treated films. However, when it is realized that these measurements are performed in a nitrogen gas environment with the ZnO approach/pull-off cycle performed numerous times, and that the immersion studies likely involve an irreversible condensation reaction, the discrepancy is reasonable.

5. Conclusions

These results demonstrate that ozone treatment of PP films and fibers oxidizes their surfaces, forming hydroxyl, carbonyl, and carboxylic acid groups. The latter are especially effective at forming covalent bonds to hydroxyl-covered ZnO nanoparticles, presumably by a condensation reaction. Heating-in-air and immersion in chromic acid, which are competing oxidation treatments, are not as effective at leading to ZnO nanoparticle adhesion, apparently due to the formation of hydroperoxide species that cannot effectively bind to the nanoparticles. While AFM force-distance measurements, performed in dry nitrogen, between oxidized surfaces and a ZnO particle are correlated to the oxidation of the surface, they are not a reliable predictor of nanoparticle adhesion to oxidized PP surfaces. Overall, the studies in the paper demonstrate that moderate ozone treatment is an effective way to attach metal oxide nanoparticles to PP surfaces. Ozone generators are used for a variety of industrial process, including in the semiconductor industry, and the pretreatment of fibers and fabrics prior to immersion in, or spray-coating with, colloidal metal oxide nanoparticles potentially may be an industrially relevant method of attaching them to polymer-based materials.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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